
ConnectomeQuest: Certificate-Carrying Interaction under Costly Partial Observation

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Abstract

1 Agents acting under partial observation should not treat an inferred state as evi-
2 dence that an answer or action is supported. We introduce CONNECTOMEQUEST,
3 a claim–receipt–certificate interface that separates solution existence, charged in-
4 formation acquisition, and independent verification. Under matched access on
5 H01, MANC, and HemiBrain, no candidate ordering dominates: random branch
6 diversification is strongest on H01, whereas an Attribute second-hop ordering im-
7 proves one held-out HemiBrain comparison by 0.0213 (two-comparison Bonferroni
8 95% interval [0.0030, 0.0397]). Replacing only the target predicate while holding
9 the anchor, wiring, controller, checkpoint, and random seed fixed raises visible
10 target prevalence from 0.0060 to 0.1476 on MANC and from 0.0191 to 0.1535
11 on HemiBrain, demonstrating strong sensitivity of acquisition to semantic target
12 granularity. In separate audits, a policy-independent verifier accepts all 22,662
13 unmodified nonempty certificates and rejects all 13,800 constructed mutations. In
14 RGB UnlockPickupDist, a prospectively frozen CPU confirmation on 221 eligible
15 long-plan layouts finds identical authorization decisions while dependency-indexed
16 validation reduces validation-and-replanning CPU time by 96.7% (layout-bootstrap
17 95% interval [96.4%, 96.9%]) relative to complete validation under four or eight
18 irrelevant receipt withdrawals; this endpoint includes index costs but excludes
19 perception, simulation, and action execution. After final support loss, checking
20 methods make zero unsupported authorizations, whereas unchecked reuse makes
21 511 of 513. Together, these results establish an access-controlled neuro-symbolic
22 protocol for acquiring, validating, and selectively revising evidence across graph
23 and RGB agents.

24 1 Introduction

25 An agent using a partially observed semantic map must distinguish what it has observed, what it
26 infers, and which acquired facts support its next decision. We organize budgeted semantic search
27 around three questions: whether a solution exists, what acquiring its evidence costs, and whether
28 another component can verify the resulting certificate. We call an interaction certificate-carrying
29 when every accepted answer is accompanied by a certificate assembled from environment-issued
30 receipts and accepted by a separately trusted verifier under the current query and episode context.

31 An observation can change confidence without improving affordable continuation actions: informa-
32 tion, decision value and witness validity are distinct quantities.

33 The principal contribution of CONNECTOMEQUEST is a certificate-carrying interaction protocol
34 and its evaluation procedure; learned rankers and symbolic planners are replaceable instantiations,
35 not the claimed protocol contribution. For any proposed output u , the task defines a finite set $\mathcal{E}(u)$
36 of required claims. Acceptance requires an active receipt supporting every required claim under a

Table 1: Notation used in the formal contract.

Symbol	Meaning
$G_d = (\mathcal{V}_d, \mathcal{R}_d, E_d, w_d)$	dataset graph and nonnegative edge weights
V, B, H	visibility-page size, action budget, and horizon
$q = (a, y); \omega = (m, x)$	query and graph witness
$\tau; P; \chi$	receipt; certificate; trusted verification context
$\mathcal{E}(u)$	finite set of required claims for output u
$s_t; z_t$	environment state and controller state
$\mathcal{K}_\pi; \pi$	controller instantiated with candidate-ordering policy π
$D_i \subseteq \mathcal{I}; K_i \subseteq \mathcal{K}$	receipt and semantic-key dependencies of plan step i
$\Delta_t = (W_t, N_t); \mathcal{J}(\Delta_t)$	typed evidence update; affected plan positions
$\Gamma_d(s, q)$	admissible API actions at environment state s
$\tilde{S}(\mathcal{K}_\pi; d, \mathcal{Q}_d, B, H)$	finite-sample certificate-success rate

37 trusted context. The graph verifier accepts final answers; the RGB verifier authorizes actions under
 38 the same typed interface. The graph benchmark then measures how semantic targets and acquisition
 39 rules affect search independently of checking; the RGB study instantiates the same required-claim
 40 interface with separate policies and predicates.

41 Connectomes provide a controlled setting for this problem: weighted synaptic wiring is combined
 42 with cell-type, region, and layer annotations. HemiBrain covers the adult fly central brain [15],
 43 MANC the male fly ventral nerve cord [18], and H01 a human cortical sample [16]. Their different
 44 degree, weight, and annotation distributions test whether an action-selection policy transfers across
 45 reconstructions and semantic schemas.

46 Full-graph query-answering and relational-search methods learn geometric or path representations
 47 [5, 8, 14]; our protocol instead charges observation and requires acquired evidence. Costed evaluation
 48 and value-of-observation planning study which observation to buy [6, 9, 12]. In embodied settings,
 49 active semantic mapping and task–motion planning couple perception, state estimation, and action
 50 constraints [1, 10]. Truth maintenance, incremental execution, and plan repair use dependency
 51 structure to localize revision [4, 7, 17, 19]. Provenance and proof-carrying code distinguish derivation
 52 records from consumer checks [11, 13]. We combine these established ideas in a controlled acquisition
 53 benchmark rather than claim a new general provenance formalism or universal planner.

54 The complete graph remains private to the environment. An agent pays for observation and traversal,
 55 then submits acquired evidence before its budget expires. This isolates action-conditioned observation,
 56 precondition enforcement and limited acquisition, also relevant to embodied semantic maps. The
 57 graph abstraction excludes low-level locomotion and manipulation; exact ground truth supports
 58 controlled acquisition and certificate audits.

59 Our contributions are:

- 60 • an explicit certificate-carrying interaction interface and a three-connectome benchmark for budgeted
 61 acquisition of semantic witnesses;
- 62 • a controller-matched held-out evaluation of six observable orderings, plus a paired target-granularity
 63 intervention that holds wiring fixed;
- 64 • a policy-independent verifier and mutation audit that isolate certificate integrity from policy success;
- 65 • a prospectively frozen RGB study showing that dependency-indexed validation preserves complete-
 66 validation decisions while reducing specified validation-and-replanning CPU time on long plans
 67 with sparse updates and preventing unsupported cached actions.

68 2 Active semantic connectome reasoning

69 **Common contract.** For task instance r , let $\mathcal{T}_r$ be its receipt space and $\mathcal{X}_r$ its trusted-context space.
 70 Define $\mathcal{C}_r = (\mathcal{U}_r, \mathcal{F}_r, \mathcal{E}_r, \text{Issue}_r, \text{Verify}_r)$, where $\mathcal{U}_r$ is the output space, $\mathcal{F}_r$ is the set of atomic
 71 claims, and $\mathcal{E}_r : \mathcal{U}_r \rightarrow \mathcal{P}_{\text{fin}}(\mathcal{F}_r)$ maps each output to its finite set of required claims. Issuance is the
 72 trusted relation $\text{Issue}_r \subseteq \mathcal{X}_r \times \mathcal{F}_r \times \mathcal{T}_r$; $(\chi, f, \tau) \in \text{Issue}_r$ means that the environment issued receipt
 73 τ for claim f in context χ . Each context also specifies the currently active receipt set $\text{Act}_r(\chi) \subseteq \mathcal{T}_r$.
 74 We say that τ supports f under χ exactly when $\tau \in \text{Act}_r(\chi)$ and $(\chi, f, \tau) \in \text{Issue}_r$. The verifier has
 75 type $\text{Verify}_r : \mathcal{U}_r \times \mathcal{T}_r^* \times \mathcal{X}_r \rightarrow \{0, 1\}$, where $\mathcal{T}_r^*$ is the set of finite receipt sequences. It accepts

76 exactly when every required claim has support in the submitted certificate:

$$\text{Verify}_r(u, P; \chi) = 1 \iff \forall f \in \mathcal{E}_r(u) \exists \tau \in P : \tau \text{ supports } f \text{ under } \chi.$$

77 The graph and RGB tasks below instantiate this interface; their policies, perception models, planners,
78 and required-claim definitions are task-specific.

79 **Graph, visibility, and query.** For dataset d , let $G_d = (\mathcal{V}_d, \mathcal{R}_d, E_d, w_d)$ be a directed weighted
80 multirelational graph, with $E_d \subseteq \mathcal{V}_d \times \mathcal{R}_d \times \mathcal{V}_d$ and $w_d : E_d \rightarrow \mathbb{R}_{\geq 0}$. Wiring uses relation
81 $p = \text{presynaptic_to}$; semantic evidence uses the dataset-specific relation ρ_d : has_type for the
82 fly graphs and in_cortical_layer for H01. For node u , relation r , page size V , and offset h , the
83 visibility operator $\text{Vis}_{d,r,V,h}(u)$ returns entries $h+1, \dots, h+V$ from the outgoing triples, sorted
84 by decreasing weight and then numeric destination ID. Write $\text{Vis}_{d,r,V}(u) = \text{Vis}_{d,r,V,0}(u)$. Each
85 subsequent $\text{INSPECT}(u, r)$ advances h for (u, r) ; changing focus does not.

86 For anchor neuron a and semantic target y , define $q = (a, y)$ and

$$A_d(a, y) = \{x \in \mathcal{V}_d : \exists m \in \mathcal{V}_d, (a, p, m), (m, p, x), (x, \rho_d, y) \in E_d\}. \quad (1)$$

87 A graph witness is $\omega = (m, x)$, with answer projection $\text{ans}(\omega) = x$, and its required claims are

$$\mathcal{E}_d(q, \omega) = \{(a, p, m), (m, p, x), (x, \rho_d, y)\}. \quad (2)$$

88 The task is to return x with acquired evidence for every member of $\mathcal{E}_d(q, \omega)$. Degenerate paths
89 are allowed only when every displayed directed edge occurs. Queries are generated at $V_{\text{gen}} = 32$
90 with one witness in the relevant first pages; $V < 32$ is therefore a visibility stress test without that
91 guarantee.

92 **Receipts and verification.** A graph receipt is $\tau = (\kappa, e, h_d, \text{id}(q), \eta, t, \iota)$, containing polarity
93 $\kappa \in \{+, -\}$, triple e , graph digest h_d , query identifier, episode nonce η , issuance step t , and receipt
94 identifier ι . The trusted context is $\chi = (h_d, q, \eta, t_{\text{cur}}, \mathcal{R})$, where registry $\mathcal{R}$ is a partial map from
95 issued identifiers to canonical payload digests. For a fixed canonical serialization and digest function
96 H , define $\text{ValidReceipt}(\tau; \chi) = 1$ exactly when its graph, query, and episode bindings match χ ,
97 $0 \leq t \leq t_{\text{cur}}$, and its digest equals $\mathcal{R}(\iota) = H(\text{payload}(\tau))$; an unregistered identifier is invalid. For
98 proposed answer x , the submission contains a graph witness $\omega = (m, x)$. A graph certificate P is
99 a finite receipt list, and $\text{Verify}_{d,q}(\omega, P; \chi) = 1$ exactly when P contains a valid positive receipt for
100 every member of $\mathcal{E}_d(q, \omega)$. Verification does not read G_d but is conditional on issuer-attested receipt
101 truth and the trusted registry.

102 **Proposition 1** (Conditional graph-certificate soundness). *If G_d is immutable and the issuer marks
103 a receipt positive only for an edge in E_d , then $\text{Verify}_{d,q}(\omega, P; \chi) = 1 \Rightarrow \text{ans}(\omega) \in A_d(a, y)$.
104 The accepted receipts instantiate Eq. (1). This does not imply certificate completeness, third-party
105 authentication, or correctness of a learned RGB interpretation.*

106 **Controlled process and controller.** The environment is the deterministic partially observed con-
107 trolled process

$$\mathcal{M}_{d,q} = (\mathcal{S}_{d,q}, \mathcal{A}_d, \mathcal{O}_d, T_{d,q}, \Omega_{d,q}, c_{d,q}, B, H, \mu_{0,d,q}),$$

108 where $T_{d,q} : \mathcal{S}_{d,q} \times \mathcal{A}_d \rightarrow \mathcal{S}_{d,q}$ is total, $\Omega_{d,q} : \mathcal{S}_{d,q} \rightarrow \mathcal{O}_d$ is the observation map, and $c_{d,q} :$
109 $\mathcal{S}_{d,q} \times \mathcal{A}_d \rightarrow \mathbb{R}_{\geq 0}$ is the charged cost, B the cost budget, H the horizon, and $\mu_{0,d,q}$ the initial-state
110 law. Admissible actions form $\Gamma_d(s, q) \subseteq \mathcal{A}_d$; an inadmissible attempt yields an API-rejection
111 transition and still incurs its stated cost. Environment state s_t contains the hidden graph, focus, issuer
112 state, pagination offsets, step, and remaining budget. Controller state z_t contains policy memory and
113 derived frontier state. For a candidate-ordering policy π , controller $\mathcal{K}_\pi$ and a fixed random tape ξ , let
114 $F_{\mathcal{K}_\pi}$ denote the controller-state update. They induce

$$\begin{aligned} o_t &= \Omega_{d,q}(s_t), & a_t &= \mathcal{K}_\pi(z_t, o_t; \xi), \\ s_{t+1} &= T_{d,q}(s_t, a_t), & z_{t+1} &= F_{\mathcal{K}_\pi}(z_t, o_t, a_t, \Omega_{d,q}(s_{t+1}); \xi). \end{aligned}$$

115 **Actions and evaluation.** The evaluated controller uses `INSPECT`, `TRAVERSE`, `CHECKEDGE`,
116 `BACKTRACK`, and `SUBMIT`. `CHECKEDGE` is admissible only for a semantic triple (x, ρ_d, y) when x
117 is the focus and an acquired two-hop wiring path ends at x . Every attempted non-submit action costs
118 one; `SUBMIT` costs zero and consumes one step. The primary controller fixes $V = 32$, $B = H = 64$,
119 and at most four answer checks per opened middle node.

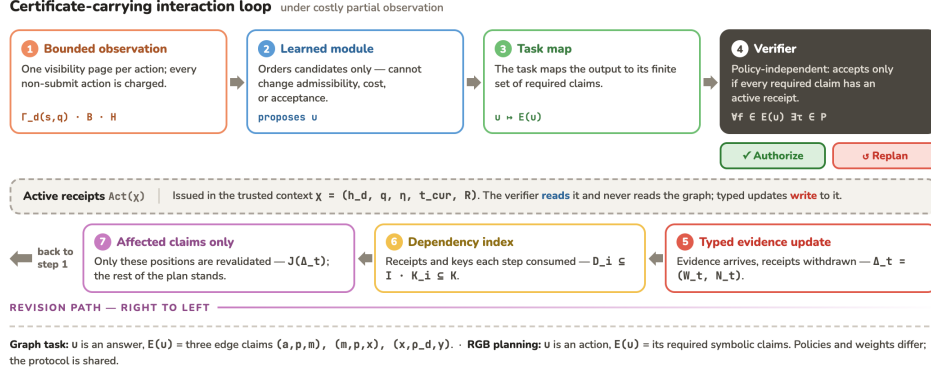


Figure 1: Certificate-carrying interaction loop. Graph and RGB tasks share receipt-backed authorization and dependency-indexed revision, but not policies or task semantics.

For fixed query set $\mathcal{Q}_d = \{q_1, \dots, q_n\}$, let $\mathcal{L}_\pi$ be the fixed evaluation-seed set for policy π , with a singleton set for a deterministic policy. Let $Y_{i\ell} = 1$ iff controller $\mathcal{K}_\pi$ submits within B, H an answer whose graph certificate is accepted for q_i under seed $\ell \in \mathcal{L}_\pi$, and 0 otherwise. The reported statistic is

$$\widehat{S}(\mathcal{K}_\pi; d, \mathcal{Q}_d, B, H) = \frac{1}{n} \sum_{i=1}^n \frac{1}{|\mathcal{L}_\pi|} \sum_{\ell \in \mathcal{L}_\pi} Y_{i\ell}.$$

It is a finite-sample certificate-success rate conditional on the named controller, queries, checkpoints, and seeds, not an unspecified population probability.

3 Benchmark implementation

3.1 Learned candidate ordering

The Attribute scorer receives no numeric entity IDs or entity embeddings. Its two MLP heads rank each supplied hop-specific candidate list from 31 observable coordinates: acquired counts and weights, memory/frontier flags, within-list ranks, budget state, semantic matches, and paid-probe statistics when available. It is trained only on source graphs for 20 epochs with AdamW at 3×10^{-4} . Numeric-ID tie-breaking can still change visibility and rank-derived features; complete features, loss, sample counts, and the relabeling counterexample are in the supplement.

3.2 Controller and trust boundary

The fixed controller instantiates Eq. (1) as a query automaton. It inspects the anchor, orders the visible middle nodes, traverses to a selected middle node, inspects its successors, checks legal semantic claims, and backtracks on failure. Learned scores only permute candidate lists; the controller owns focus, action masking, budget, and the required-claim set. The general graph API permits more actions than this automaton, so the reported protocol is defined by controller $\mathcal{K}_\pi$ as well as API legality.

4 Experimental design

Data. We construct compact neuron-level KGs from official releases and exclude raw EM imagery, skeletons, meshes, and per-synapse point tables; the supplement reports the exact node and edge counts by relation type. H01 annotations are official cortical-layer labels sampled at soma centroids. MANC retains neurons with a non-null curated mancType; HemiBrain retains traced, uncropped neurons exported through neuPrint [3]. Inverse wiring edges are materialized. We use publicly released datasets under their respective research licenses; the supplement records source versions and access metadata.

149 **First-page target prevalence.** Define the *first-page two-hop candidate union*

$$X_{d,V}^{\text{fp}}(a) = \{x \in \mathcal{V}_d : \exists m, (a, p, m) \in \text{Vis}_{d,p,V}(a), (m, p, x) \in \text{Vis}_{d,p,V}(m)\}.$$

150 This set is computed offline by opening the first continuation page for every middle node disclosed
 151 on the anchor’s first page. It is a dataset diagnostic, not an agent observation, and no evaluated policy
 152 receives it. For query (a, y) , its visible target prevalence is

$$s_{d,V}(a, y) = \frac{|A_d(a, y) \cap X_{d,V}^{\text{fp}}(a)|}{\max\{1, |X_{d,V}^{\text{fp}}(a)|\}}.$$

153 A smaller value means that the target selects a smaller fraction of this offline candidate union. Target
 154 counts and answer-set sizes are accompanying statistics, not alternative definitions. Test queries use
 155 all seven H01 layer targets, 1,945 MANC type targets among 3,893 graph types, and 2,278 HemiBrain
 156 type targets among 5,554 graph types. Mean answer-set sizes are 45.5, 7.3, and 20.7, respectively.
 157 Cross-graph differences also include topology, weights, and annotation coverage; we do not attribute
 158 them to target prevalence alone.

159 **Splits and evaluation blocks.** Neuron anchors are assigned to train, validation, or test by seeded
 160 cryptographic hash; semantic targets are shared ontology nodes, not held-out neurons. For each
 161 leave-one-connectome-out direction, the Attribute and Recurrent policies train on the other two
 162 graphs with seeds 17, 29, and 43 and receive no entity embeddings. Hyperparameters and selector
 163 checkpoints use source validation only.

164 The main comparison is the prospectively frozen held-out-pair evaluation in Table 2. It holds
 165 the query automaton, visibility operator, budget, action restrictions, verifier, and query set fixed
 166 while changing only candidate ordering. Its controller ranks the complete visible first-hop frontier
 167 without a paid probe. The 5,000-pair target-coarsening block holds the anchor, wiring, visibility rule,
 168 controller, policy checkpoint, and random seed fixed while replacing only the target predicate; because
 169 that predicate changes the query, subsequent actions may differ. This is an explicitly exploratory
 170 mechanism audit.

171 **Baselines and statistical unit.** All main-table policies use observable information only. Random
 172 applies a deterministic permutation keyed by run seed, query ID, current node, and candidate ID. The
 173 Recurrent baseline uses the same 31 observed features and source-graph decision snapshots as the
 174 Attribute scorer, but trains an LSTM ranker with REINFORCE; its state resets at each ranking call. It
 175 is therefore an access-matched recurrent control, not a reproduction of full-graph MINERVA. The
 176 supplement gives architectures, training volumes, and hyperparameters.

177 Certificate success is averaged over 1,000 fixed query IDs and, for stochastic policies, three fixed
 178 training/action seeds. After an audit found repeated first anchors, we replaced query-wise resampling
 179 by a paired cluster bootstrap; this post-outcome correction changes uncertainty estimates, not policies
 180 or point estimates. It first averages seed outcomes within query, then resamples first-anchor clusters
 181 10,000 times and retains every query in each sampled cluster. H01 has 837 clusters, MANC 807,
 182 and HemiBrain 806 (maximum cluster size four). The two planned fly-graph contrasts between
 183 Expert selector and Weight use Bonferroni-adjusted 97.5% intervals. H01 and all other contrasts
 184 are descriptive. The correction accounts for repeated anchors but remains conditional on fitted
 185 checkpoints, fixed seeds, the controller, and these graphs.

186 5 Results

187 5.1 Held-out evaluation under fixed access

188 We fixed the controller, page size $V = 32$, budget and horizon $B = H = 64$, candidate cap, policy
 189 checkpoints, and two fly-graph comparisons before evaluating 1,000 previously unused query pairs
 190 per graph. The queries share no anchor–target pair with earlier evaluation sets. Because first anchors
 191 repeat, confidence intervals resample first-anchor clusters. The three action/model seeds are fixed
 192 computational replications; the intervals are conditional on these seeds and on this query population.

193 Weight orders candidates by observed synaptic weight. Random uses a query- and seed-keyed random
 194 permutation. Recurrent is an access-matched LSTM policy trained with REINFORCE. Attribute

is the two-head listwise MLP in Section 3. Expert selector chooses one of the observable policies from source-graph features before acting. Random→Attribute randomizes only the first hop and uses Attribute ordering at the second. Every policy receives the same observations, action costs, and certificate-verification rule.

Table 2: Prospectively frozen evaluation on 1,000 previously unused query pairs per graph. Entries are certificate success/unconditional charged actions, averaged across seeds 17/29/43 for stochastic policies; Weight is deterministic. All policies use the same frozen controller and observations.

Graph	Weight	Random	Recurrent	Attribute	Expert selector	Random→Attribute
H01	0.4090/42.63	0.5100/39.82	0.4840/40.64	0.4117/41.27	0.3920/43.14	0.4017/42.05
MANC	0.0990/60.18	0.1050/60.11	0.1030/60.01	0.1107/59.96	0.1120/60.03	0.1003/60.19
HemiBrain	0.1270/59.52	0.1347/59.23	0.1437/58.86	0.1343/58.91	0.1483/58.83	0.1367/58.86

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No policy dominates on all three graphs. Relative to Weight, Expert selector changes certificate success by +0.0213 on HemiBrain (two-comparison Bonferroni 95% interval [0.0030, 0.0397]), +0.0130 on MANC ([−0.0094, 0.0360]), and −0.0170 on H01 (descriptive nominal interval [−0.0295, −0.0048]). Random is strongest on H01, whereas Expert selector is strongest on HemiBrain. A matched hop-attribution audit shows that replacing the selector’s first hop by Weight leaves all 9,000 outcomes unchanged. Thus the observed HemiBrain difference is attributable to second-hop ordering, not to the first-hop selector. Accordingly, the selector is an evaluated component rather than the primary contribution.

Raw action means can reward early failure. Outcome-conditioned aggregation shows the opposite: unsuccessful episodes average about 51 charged actions on H01 and 63 on MANC and HemiBrain. The controller reaches its 63-action exploration limit in 29–37% of H01 episodes, 89–90% of MANC episodes, and 85–87% of HemiBrain episodes. The supplement gives every policy; these data identify acquisition exhaustion rather than low-cost failure.

212 5.2 Target granularity controls acquisition difficulty

Let Y_d be the native target set and $\bar{Y}_d = \{\bar{y}_1, \dots, \bar{y}_7\}$ a disjoint set of synthetic coarse targets. A fixed map $g_d : Y_d \rightarrow \bar{Y}_d$ replaces each semantic edge (x, ρ_d, y) by $(x, \bar{\rho}_d, g_d(y))$. Consequently $A_d(a, y) \subseteq \bar{A}_d(a, g_d(y))$. This set inclusion is elementary; the experiment measures its effect on paired anchors under a fixed visibility rule, controller, policy checkpoint, and random tape. The target predicate, and hence the realized actions, may differ.

We apply the transformation to 5,000 paired MANC and HemiBrain instances while holding wiring, anchor, pairing identifier, policy checkpoint, and random seed fixed. Visible target prevalence

Table 3: Paired label-coarsening audit, 5,000 original native queries per fly graph. Every entry is native/coarse. Visible target prevalence is the relevant fraction among first-page two-hop candidate nodes; other columns are Success(64), averaged over the original action/model seeds. No new model fitting.

Graph	Visible target prevalence	Weight	Random	Attribute
MANC	0.0060/0.1476	0.0972/0.8226	0.1093/0.9036	0.1088/0.8644
HemiBrain	0.0191/0.1535	0.1290/0.7858	0.1765/0.8887	0.1373/0.8167

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increases from 0.0060 to 0.1476 on MANC and from 0.0191 to 0.1535 on HemiBrain; success rises substantially for all three reported policies. Thus semantic target granularity materially changes acquisition difficulty in this controlled intervention; it is not evidence that one ordering algorithm alone solves the native regime. A privileged visible-witness ordering solves every generated query within 16 actions, confirming that the gap is observable acquisition rather than answer non-existence.

225 5.3 Certificate verification

The policy-independent standard-library verifier sees the public query, proposed witness, certificate, and trusted issuance context, but no graph or answer set. It accepts all 22,662 unmodified nonempty

submissions from 60,000 graph-search episodes; empty submissions are not certificates. In a separate diagnostic it rejects all 13,800 constructed binding and structural mutations. This tests policy-independent logical support under a trusted issuer, not search quality or an attack-rate bound; class-by-graph counts and the trusted-registry assumption are in the supplement.

6 RGB semantic-map evaluation

BabyAI UnlockPickupDist [2] instantiates the same claim–receipt–certificate interface. Every method receives the same egocentric 7×7 RGB image, position relative to the initial pose, exact cardinal direction, inventory type, mission string, and last-action outcome. A frozen tile classifier converts the RGB image into semantic-map claims, each linked to the corresponding immutable image receipt. For a proposed action a , $\mathcal{E}_t(a)$ is its finite set of required symbolic claims. Authorization requires active receipt support for every claim in $\mathcal{E}_t(a)$. A receipt proves image acquisition, not the physical truth of the learned interpretation.

Dependency-indexed plan validation. Let $\mathcal{I}$ be the receipt-identifier set and $\mathcal{K}$ the set of semantic update keys. At decision time t , the deterministic planner’s retained suffix is $\Pi_t = (a_{c_t}, \dots, a_L)$. Let $A_t \subseteq \mathcal{I}$ be the active receipts and let θ_t be the current semantic-key valuation. For every $i \geq c_t$, define $\text{Supp}_i(A_t, \theta_t) \in \{0, 1\}$ to indicate whether every required claim of a_i has support from an active receipt.

The recorded set $D_i \subseteq \mathcal{I}$ is *receipt-complete* when the following implication holds for all A, A' and θ : if A and A' differ only in receipt r and $\text{Supp}_i(A, \theta) \neq \text{Supp}_i(A', \theta)$, then $r \in D_i$. The recorded set $K_i \subseteq \mathcal{K}$ is *key-complete* when it contains every semantic key whose value is read either by Supp_i or by the planner when constructing a_i and its required-claim set. A changed key in K_i therefore marks the retained suffix stale even if its currently cached support value remains one. For an update from (A_t, θ_t) to (A_{t+1}, θ_{t+1}) , let $W_t = A_t \triangle A_{t+1}$ and let N_t contain exactly the keys whose values changed; write $\Delta_t = (W_t, N_t)$. The affected positions are

$$\mathcal{J}(\Delta_t) = \{i \geq c_t : D_i \cap W_t \neq \emptyset \text{ or } K_i \cap N_t \neq \emptyset\}.$$

The dependency index stores the exact inverse maps $L_{\mathcal{I}}(r) = \{i : r \in D_i\}$ and $L_{\mathcal{K}}(k) = \{i : k \in K_i\}$.

Proposition 2 (Coverage and authorization). *If D_i is receipt-complete, K_i is key-complete, and both inverse maps are exact, then $\mathcal{J}(\Delta_t)$ contains every suffix position whose support value changes and every position whose recorded planner-read key changes. Consequently, a controller that revalidates those positions, replans after a recorded planner-read change, and authorizes the next action only with current support never authorizes that checked action after its final active support is withdrawn. The conclusion is relative to the trusted issuer and symbolic action model; it does not establish physical correctness.*

Proposition 3 (Selective-validation equivalence). *Suppose complete suffix validation and dependency-indexed validation receive the same retained suffix, active receipts, semantic valuation, update, and deterministic planner, and an exact cached pre-update support vector. Both procedures use the same retention rule: retain exactly when all post-update support values equal one and $N_t \cap \bigcup_{i=c_t}^L K_i = \emptyset$; otherwise invoke the common planner. Under the completeness assumptions of Proposition 2, they therefore return the same next authorization decision. This statement does not compare either procedure with eager replanning after a change outside the recorded read dependencies that could make an unrelated alternative plan preferable.*

Corollary 1 (Amortized location work). *Within one retained-plan epoch with fixed cursor c , let $Z = \sum_{i=c}^L (|D_i| + |K_i|)$ at index construction. With expected-constant-time hash lookup, construction followed by M updates has location work*

$$O\left(Z + \sum_{t=1}^M \left[|W_t| + |N_t| + \sum_{r \in W_t} |L_{\mathcal{I}}(r)| + \sum_{k \in N_t} |L_{\mathcal{K}}(k)| \right] \right).$$

Complete suffix validation instead visits $M(L - c + 1)$ positions. Across multiple retained-plan epochs, both costs sum over epochs. Predicate checks, index maintenance, and planning after rejection are additional costs and are reported separately in the experiment.

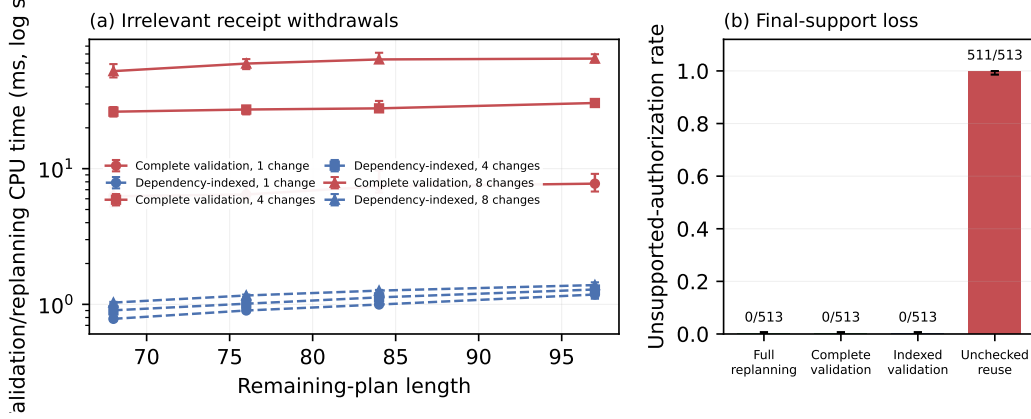


Figure 2: Two distinct interventions under matched RGB access. (a) In the primary long-plan population with four or eight irrelevant receipt withdrawals, dependency-indexed and complete retained-plan validation agree on all 1326 paired decisions, while indexing reduces validation-and-replanning CPU time by 96.7% (layout-bootstrap 95% interval [96.4%, 96.9%]); curves also show the one-withdrawal condition descriptively. (b) After final-support loss, every checking method prevents unsupported authorization, whereas unchecked reuse does not. Binomial bars are exact 95% intervals.

275 All four methods share the planner, action model, RGB observations, and update schedule. A method-
 276 independent rule retains 513 of 800 generated layouts, yielding 3,078 paired cells; every stored trace
 277 replays exactly.

278 The prospectively frozen primary population contains 221 layouts with at least 64 remaining ac-
 279 tions and four or eight distinct irrelevant receipt withdrawals, yielding 1326 paired timing cells.
 280 Dependency-indexed and complete retained-plan validation agree on 100.0% of authorization de-
 281 cisions. We define *validation-and-replanning CPU time* as index construction, index maintenance,
 282 required-claim validation, and planner calls; it excludes CNN inference, simulator stepping, and
 283 action execution. Dependency indexing reduces this endpoint by 96.7% relative to complete valida-
 284 tion (layout-bootstrap 95% interval [96.4%, 96.9%]). Panel (a) additionally shows the prospectively
 285 specified one-update condition descriptively.

286 In a separate final-support-loss intervention, full replanning, complete validation, and dependency-
 287 indexed validation make zero unsupported authorizations in 513 episodes; unchecked reuse makes
 288 511. Thus the efficiency result concerns sparse irrelevant changes, while the safety result tests removal
 289 of the final support for the current action. Neither comparison asserts an advantage in physical action
 290 count or perception accuracy.

291 7 Discussion

292 The experiments separate three facts. Target granularity changes visible target prevalence and witness
 293 acquisition under fixed wiring and budgets. No ordering is uniformly best: random diversification
 294 is competitive on two graphs, while learned second-hop ordering helps on HemiBrain. Certificate
 295 validity is independent of search quality: weak policies may return no certificate, and success requires
 296 active receipt support for every required claim.

297 The RGB study isolates evidence management. On long plans under sparse updates, dependency
 298 indexing preserves complete-validation decisions while reducing validation-and-replanning CPU
 299 time; unchecked reuse fails after final support loss. It does not evaluate reinforcement learning,
 300 action efficiency, perceptual correctness, or transfer of task-specific policies, planners, and claims.
 301 Under trusted issuance, complete dependencies, and declared updates, Proposition 3 equates the two
 302 validation procedures, not either with global replanning; it neither authenticates receipts nor detects
 303 issuer error (anonymous code and artifacts).

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